Markdown Buddy

BDA400 Assignment 1 | Data Science Tools and Techniques

Vikas Palan

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AI Assistance Declaration

AI Assistance Declaration: OpenAI Codex (model version unavailable), 27 October 2026, assisted with documentation review, revision, formatting, and packaging. The supplied example attributes earlier drafting to Grok (xAI); its version, prompts, and prior verification are unconfirmed. Available prompts and key responses are recorded in AI_Prompts_Appendix.md. No analysis calculations were performed in this revision. Student verification, R execution, R Markdown knitting, and GitHub preview/upload remain pending. This statement does not attest that Vikas Palan performed those actions.

Submission draft: review required

This report summarizes the revised documentation for the supplied synthetic customer-purchase example. It includes no asserted analysis results. The R script and CSV are retained unchanged. Student review, execution, knitting, and publication remain outstanding.

Repository destination

https://github.com/vikpalan-cdi/markdown-buddy-vikaspalan

The user provided this destination. No GitHub upload or LMS submission was performed.

Deliverables

README.md; analysis_documentation.Rmd; Reflection.md; AI_Prompts_Appendix.md; Verification.md; supplied R/analysis.R and data/synthetic_customers.csv; original license; upload steps; ZIP archive.

01 / README design

Documentation aligned with the actual script

Purpose and scope

The script inspects synthetic purchases, prepares dates and age groups, summarizes spending by region and category, and displays a regional boxplot. It does not analyze age-group differences or save tables and plots by default.

Planned README sections

- Overview: project purpose, scope, and current status.

- Installation and example code: dependencies, project-root paths, and execution instructions.

- Data dictionary: fictional fields, formats, and unspecified currency.

- Outputs and limitations: actual objects and constraints on interpretation.

- Documentation approach, license, and AI assistance record.

Comparison with ggplot2

The ggplot2 README uses Overview, Installation, and Usage. This project follows that useful reader sequence, then adds data and verification details specific to a small analysis. Package badges and development claims were omitted because they would not describe this example.

Reproduction commands

Install tidyverse, lubridate, rmarkdown, and knitr. From the project root, run source("R/analysis.R"). To knit the walkthrough, run rmarkdown::render("analysis_documentation.Rmd") or use Knit in RStudio. These are instructions, not a record of successful execution.

02 / R Markdown documentation

One document for one supplied R script

Structure and inputs

The single .Rmd includes YAML metadata, an assistance statement, labeled executable R chunks, Purpose, Inputs, a step-by-step walkthrough, Outputs, and limitations. It reads data/synthetic_customers.csv relative to the project root. Its walkthrough reproduces the supplied script without also sourcing it, avoiding duplicate execution.

Outputs

The documented outputs are console inspection, region_summary, category_summary, and a displayed boxplot. Optional plot export in the standalone script remains commented out and requires an output directory before use.

Limitations found during review

- West is paired with Electronics, East with Clothing, and Central with Home. Separate region/category effects cannot be inferred.

- n_customers counts rows rather than distinct identifiers.

- Age groups are created but not analyzed; missing ages fall into the final age branch.

- A dollar display label does not establish a currency.

- Small synthetic groups do not justify real customer or causal conclusions.

Execution status

No Rscript executable was found on the checked command path or in the standard installation folder. R execution and knitting were not attempted. This PDF was generated as a document summary; it is not a knitted R analysis report.

03 / Verification and refinement

Completed assistant checks and remaining student work

Changes from the source example

- Replaced documentation placeholders with Vikas Palan and the existing repository URL.

- Removed the inherited numerical table rather than present it as verified output.

- Removed unsupported claims about completed student previewing and knitting.

- Aligned scope with the script and documented limitations.

- Preserved script, CSV, and license bytes; added a separate verification record.

Assistant checks

Source inspection, primary-documentation comparison, local link checks, Markdown fence balance, unique R chunk labels, CSV schema, and copied-file integrity were checked. The summary PDF was rendered for visual layout review. These do not establish student validation or successful R execution.

Required student work

Run the script, knit the .Rmd, inspect the rendered HTML, and record actual R/package versions and warnings. If numerical results are reported, complete the specified primary calculation in Sheets, independent cross-check, and self or peer spot-check. No analysis calculations were performed in this revision.

Publication

Browser control stopped because it could not confidently identify the current Chrome URL. Nothing was uploaded. Follow Upload_Steps.md to publish reviewed files, inspect the README in GitHub, and submit the final reviewed PDF through the LMS.

04 / Reflection

Draft responses to the questions in the supplied ZIP

The instruction PDF refers to three questions but does not print them. The supplied ZIP provides the headings used below. Confirm them against the course and personalize the full Reflection.md after completing actual review.

1. AI strengths and weaknesses

AI can organize practical installation and usage information and improve formatting consistency. The example demonstrates a risk: fluent documentation can describe analysis the script does not perform or claim validation without evidence. Source-based review is essential; creating age groups does not establish an age-group analysis, and a static table is not proof of execution.

2. Validation and refinement

The assistant compared the source files with their descriptions and consulted GitHub, ggplot2, and R Markdown documentation. The revision removes unverified results, corrects placeholders, and explains output behavior and data limitations. Student execution, knitting, cross-checking, and previewing remain pending. The final reflection must distinguish personal actions from assistant review.

3. Documentation and ethical AI use

Effective documentation states observable behavior: displayed versus saved output, row counts versus distinct customers, and synthetic descriptions versus supported findings. Responsible use records assistance and distinguishes drafting from validation. The student still needs to understand the work, perform checks, and take responsibility for the final submission. This draft does not claim experiences that have not been confirmed.

05 / Appendix and references

Available prompt record and exact course declaration

Prompt record

AI_Prompts_Appendix.md records the available preparation request and key assistant actions. Four prompts inherited from the example are labeled unverified historical text. Suggested seed/refinement/critique prompts are labeled not yet executed. Unavailable prior Grok and ChatGPT response text has not been invented.

Exact course declaration: attestation pending

The required template is reproduced verbatim below, not as a signed or completed assertion:

AI Assistance Declaration: I used {tool} for {what}. Prompts used: {paste}. I verified outputs using {methods}. All final calculations are done by myself. I am responsible for the accuracy and originality of this work.

After completing actual checks, fill the braces truthfully and place the completed statement at the top of each submitted file. Keep YAML first in R Markdown. Do not certify unperformed verification or personal calculation.

Sources consulted, 27 October 2026

Supplied five-page BDA400 Assignment 1 instructions PDF; supplied markdown-buddy-example.zip. Provenance beyond the source file contents is unconfirmed.

GitHub Docs: Basic writing and formatting syntax. https://docs.github.com/en/get-started/writing-on-github/getting-started-with-writing-and-formatting-on-github/basic-writing-and-formatting-syntax

tidyverse/ggplot2 README: https://github.com/tidyverse/ggplot2/blob/main/README.md

R Markdown: Code Chunks. https://rmarkdown.rstudio.com/lesson-3.html